compliance with investor protection rules[856] and that multiple NBBOs would affect surveillances that detect broker-dealer market access controls deficiencies and fraud and manipulation.[857] The commenter also stated that the proposed model would make it almost impossible to monitor for locked and crossed markets and to conduct trade-through surveillance due to the multiple competing consolidator feeds.[858] In addition, the commenter stated that regulators and market participants will have different views of the market depending on the NBBO they use, which would allow bad actors to take advantage of the resulting regulatory loopholes and inefficiencies.[859] The commenter stated that multiple NBBOs could create false positives, more investigations and data requests, as well as false negatives.[860]

Additionally, one commenter stated that the multiple NBBOs that will be produced by competing consolidators and self-aggregators under the proposed decentralized consolidation model will increase costs for exchanges to enforce their rules[861] and increase the risk of market manipulation.[862] This commenter also said that regulations could be inconsistently applied due to variations in consolidated market data from different competing consolidators,[863] and that the

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856 See Joint CRO Letter at 2, 4.

857 See id. at 3.

858 See id. at 4. The commenter stated that reprogramming its systems to comply with and surveil for compliance with the locked and crossed and order protection rules would be an "extensive undertaking." Id. See also Nasdaq Letter IV at 12, 36, 49.

859 See Joint CRO Letter at 3.

860 See id. at 2, 3.

861 See Nasdaq Letter IV at 12, 37. See also Joint CRO Letter at 4.

862 See Nasdaq Letter IV at 12.

863 See id. The commenter also stated that the proposal would lead to confusion and inconsistent application if adopted. See id. at 12–13.

proposal would result in more frequent enforcement investigations, expansive data requests, and delays in concluding investigations.[864] Another commenter stated that the proposed decentralized consolidation model would hurt Rule 605 compliance because the baseline NBBO used by each market center for comparisons would vary.[865]

These issues are not new or unique to the decentralized consolidation model. As discussed earlier,[866] SRO surveillance and regulatory programs currently need to consider different sources of market data, including different "NBBOs" calculated by different market participants as well as the NBBO calculated by exclusive SIPs pursuant to Rule 600(b)(50). While there may be an increased number of data sources in a decentralized consolidation model, the issues described by these commenters are currently dealt with by SRO market surveillance and enforcement staff today. Broker-dealers have considered multiple data sources in complying with trade-through and locked and crossed markets in the context of Regulation NMS. The Commission believes that the amended definition of NBBO should not cause increased exchange enforcement costs or an increase in alerts or false positives because the surveillance and regulatory process will continue as it does today.

Currently, SROs already must decide among data sources to use for their surveillance and regulatory systems. SROs must also handle the different data sources used by market participants for regulatory compliance. Market participants may use SIP data, proprietary data, or a combination. Market participants also may calculate their own NBBOs. The data used by market participants today may differ from the data used by the SROs for their surveillance and

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[864] See id. at 37.

[865] See NYSE Letter II at 24.

[866] See supra Section III.B.8.

regulatory systems. Broker-dealers are not required to use a particular source of market data for regulatory compliance. Because broker-dealers may rely on different sources of data—data sources that may not be used by SRO regulatory staff to generate alerts—SRO investigators typically request data from broker-dealers to evaluate whether an alert has discovered an actual violation of the Federal securities laws and exchange rules. This dynamic will not change in a decentralized consolidation model. SROs may decide to use a consolidated market data product generated by a competing consolidator, or they may decide to self-aggregate data to generate consolidated market data for purposes of their surveillance and regulatory systems. SROs will not be required to purchase every consolidated feed to conduct enforcement or surveillance, just as they are not required to purchase all direct proprietary feeds used by market participants. Therefore, SROs should not incur higher enforcement costs because the regulatory process will continue to operate as it does currently.[867]

Furthermore, even if SRO surveillance and regulatory programs used the same data source as a broker-dealer, SROs would still have to request information from such broker-dealer due to potential data differences—such as geographic latencies, access options, or processing options—to evaluate whether there is a potential violation. Therefore, the Commission believes that the existence of multiple NBBOs should not present any problems that the SROs have not previously handled.

C. Competing Consolidators

The Commission believes that the introduction of competing consolidators as the entities responsible for the collection, consolidation, and dissemination of consolidated market data products will update and modernize the national market system and provide market participants

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[867] See infra Section V.C.2(d); notes 1757–1760 and accompanying text.

and investors with benefits. As noted above, the Commission received many comments that supported the goals of the decentralized consolidation model and the potential positive impacts this model would have on the provision of consolidated market data.[868] However, some comments raised concerns about the introduction of competing consolidators and their impact on the national market system, which are discussed below.

1. Definition of "Competing Consolidator" under Rule 600(b)(16)

(a) Proposal

The Commission proposed to amend Regulation NMS to introduce competing consolidators as the entities responsible for the collection, consolidation, and dissemination of consolidated market data. Under proposed Rule 600(b)(16) of Regulation NMS, a competing consolidator would be defined as "a securities information processor required to be registered pursuant to Rule 614 or a national securities exchange or national securities association that receives information with respect to quotations for and transactions in NMS stocks and generates consolidated market data for dissemination to any person."

(b) Final Rule and Response to Comments

The Commission received one comment supporting the proposed definition of competing consolidator.[869] The Commission, however, is revising the definition to reflect that competing

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868 See supra Section III.B.1; BlackRock Letter, Fidelity Letter, State Street Letter, Wellington Letter, ICI Letter, Virtu Letter, AHSAT Letter, FIA PTG Letter, IntelligentCross Letter, Committee on Capital Markets Regulation Letter, BestEx Research Letter, Wharton Letter, MEMX Letter, Clearpool Letter; T. Rowe Price Letter; Capital Group Letter; DOJ Letter; ACS Execution Services Letter; IEX Letter; Steinmetz Letter (commenting on the entire proposal); SIFMA Letter; ACTIV Financial Letter; MFA Letter; Better Markets Letter.

869 Data Boiler Letter I at 46.

consolidators are not required to generate a complete consolidated market data product.[870] Rule 600(b)(16) defines a competing consolidator as "a securities information processor required to be registered pursuant to Rule 614 or a national securities exchange or national securities association that receives information with respect to quotations for and transactions in NMS stocks and generates a consolidated market data product for dissemination to any person."

Regarding the consolidated market data that must be generated by competing consolidators, the Commission asked in the Proposing Release whether competing consolidators should be required to develop a consolidated market data product that contained all of the elements provided under the proposed definition of consolidated market data.[871] As proposed, competing consolidators would have had the flexibility to offer different products but were also required to offer a full consolidated market data product. One commenter stated that competing consolidators should not be required to include depth-of-book in the products they sell, as including such data would increase latency and complexity.[872] The commenter stated that not mandating the provision of depth-of-book data would encourage competition further and reduce data costs for all consumers.[873] Another commenter stated that the requirement that competing consolidators develop a consolidated market data product that contained all of the elements of consolidated market data, regardless of demand, along with other proposed competing

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870 See supra Section II.B.2. See also infra Section III.C.8(a)(ii) for a discussion of the revisions to the responsibilities of competing consolidators resulting from this change.

871 See Proposing Release, 85 FR at 16782.

872 See BestEx Research Letter at 2, 5. The commenter said the result would be consumers that prefer depth over latency would be able to find the right products, as would those that prefer latency over depth.

873 See id. at 5.

consolidator requirements, including registration and Regulation SCI, imposed costs on competing consolidators that would serve as a barrier to entry to potential competing consolidators.[874] Another commenter suggested that only a single competing consolidator should be obligated to provide a consolidated market data product, not all competing consolidators.[875] One commenter suggested that competing consolidators should have the flexibility to compete with proprietary data offerings and decide what products to offer to their subscribers.[876]

The Commission agrees with the comments that objected to requiring that all competing consolidators sell a product containing all of the specified elements of consolidated market data[877] because forcing higher fixed costs and a mandatory product offering across all competing consolidators potentially would make it more difficult for competing consolidators to enter the market and to tailor their services and product offerings while recouping expenses. The Commission believes that allowing competing consolidators to generate their own set of product offerings, tailored to the needs of their subscriber base or otherwise differentiated, will lower the fixed costs of those competing consolidators that elect to specialize in targeted products that do not contain all of the elements of consolidated market data. Specifically, these competing consolidators can offer such products without acquiring the underlying data and developing the technological capability necessary to calculate and consolidate all elements of consolidated market data. This reduction in fixed costs will enable these competing consolidators to offer products that do not contain all elements of consolidated market data and do so in a more cost-

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874 See NYSE Letter II at 14–15.

875 Data Boiler Letter I at 52.

876 BlackRock Letter at 2–3.

877 See BestEx Research Letter at 2, 5; NYSE Letter II at 14–15; BlackRock Letter at 2–3; Data Boiler Letter I at 52.

effective manner, which should enhance their ability to meet the specific data needs of various market participants. In addition, increased participation by more streamlined competing consolidators that operate with fewer constraints and potentially fewer fixed costs, or costs tailored to their own unique product offerings, would promote competition in the market.

The Commission recognizes that competing consolidators are the only entities that will generate and disseminate consolidated market data products in the national market system, and acknowledges the possibility that competing consolidators may not offer consolidated market data products that do not contain all of the elements of consolidated market data. However, consistent with the views expressed by a variety of market participants regarding the importance of the individual elements of consolidated market data, the Commission believes that there will be widespread demand for a product that contains all elements of consolidated market data, and particularly for the additional information included in core data.[878] Specifically, because this additional data will provide useful information that will assist in order routing and placement decisions and achieving improved executions for customer orders,[879] many broker-dealers that

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878 See supra note 112 and accompanying text (noting that multiple Roundtable panelists and commenters supported the inclusion of odd-lot information, depth of book data, and auction information); supra notes 152–153 and accompanying text (describing comments expressing support for the expansion of consolidated market data and core data); 242–244 (describing comments expressing support for the inclusion of smaller-sized orders in higher-priced stocks); 386–388 (describing comments expressing support for the inclusion of depth of book data); 462–464 (describing comments expressing support for the inclusion of auction information).

879 See Proposing Release, 85 FR at 16754 (describing how depth of book data can assist Smart Order Routers and electronic trading systems with the optimal placement of orders across markets); 16759 (explaining how information regarding the size and side of order imbalances can indicate the direction a stock's price might move, inform decisions on where to price an auction order and what order type to use, and improve execution quality). See also supra Sections II.C.2(b)–II.D.2 (explaining how aggressively priced

execute customer orders will acquire a product containing all elements of consolidated market data to compete effectively for customer business.[880] Consolidated market data includes information that the Commission believes is necessary to disseminate under the rules of the national market system and useful for a broad-cross section of market participants. Accordingly, the Commission believes that there should be demand for products containing all elements of consolidated market data even though, as adopted, Rule 614 will not require competing consolidators to offer them.

With respect to the receipt of data by competing consolidators, Rule 603(b), as adopted, requires each SRO to provide its NMS information, including all data necessary to generate consolidated market data, to competing consolidators. In accordance with Rule 603(b), competing consolidators will receive each SRO's market data that is necessary to generate consolidated market data. One commenter stated that competing consolidators, their subscribers, and self-aggregators should be permitted to receive data elements or products based on need.[881] As discussed above, competing consolidators will be permitted to develop different types of consolidated market data products based on the demands of their customers.[882] Accordingly, competing consolidators will not be required to receive all of the data content underlying consolidated market data. Rather, competing consolidators can receive the data they need to generate the consolidated market data products they decide to offer, but the Commission notes

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odd-lots will be included in core data through the definitions of odd-lot information and round lot); II.F.2; II.G.2.

880 See supra Section II.C.2(a).

881 See SIFMA Letter at 11.

882 See supra Section II.B.2 discussing the definition of consolidated market data product.

that competing consolidators that are SIPs required to be registered pursuant to Rule 614 must comply with the Vendor Display Rule when offering consolidated market data products and therefore must receive, at a minimum, the data necessary to satisfy the Vendor Display Rule.[883]

Competing consolidators will also have the ability to select for themselves, from among the options offered by the SROs, how to access the data underlying consolidated market data (e.g., with different latencies, throughput capacities, and data-feed protocols) and consider costs and complexities related to each option. The Commission believes that these provisions will provide competing consolidators with flexibility to develop their consolidated market data products business in a way that best suits their capabilities and their subscribers' needs.

2. Comments on Resiliency

The Commission received several comments with respect to the impact of competing consolidators on the resiliency of the markets.[884] Some commenters stated that competing

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883 The Vendor Display Rule requires a SIP that provides a display of quotation and transaction information with respect to an NMS stock, in the context of which a trading or order-routing decision can be implemented, to also provide a consolidated display for that stock. Rule 603(c) of Regulation NMS, 17 CFR 242.603(c). See also supra note 97; FINRA, Regulatory Notice 15-46, 1, 3 n.12 (2015) ("The exercise of reasonable diligence to ascertain the best market under prevailing market conditions can be affected by the market data, including specific data feeds, used by a firm. For example, a firm that regularly accesses proprietary data feeds, in addition to the consolidated SIP feed, for its proprietary trading, would be expected to also be using these data feeds to determine the best market under prevailing market conditions when handling customer orders to meet its best execution obligations.").

884 See Committee on Capital Markets Regulation Letter at 3; BestEx Research Letter at 1; BlackRock Letter at 5; ACS Execution Services Letter at 5; Clearpool Letter at 7–8; Cboe Letter at 25; TechNet Letter II at 2; NYSE Letter II at 24; Data Boiler Letter I at 48; Data Boiler Letter II at 1; NBIM Letter at 6; NovaSparks Letter at 1. The Commission also received one comment that urged the Commission to focus on cybersecurity in upgrading the infrastructure for market data. Temple University Letter

consolidators could add resiliency to the markets.[885] One commenter said that competing consolidators would strengthen resiliency because there would no longer be a single point of failure that could cause "stock market paralysis."[886] Similarly, another commenter said that competition could reduce the risk of a single point of failure,[887] and another commenter said that the decentralized consolidation model would eliminate the SIP as a single point of failure.[888] Two commenters said that having a "sufficient number of competing consolidators will be important to ensure resiliency and backup in the collection, consolidation, and distribution of consolidated market data."[889]

Commenters also argued that introducing competing consolidators could weaken the national market system by increasing a risk of failure,[890] or that competing consolidators would

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at 2. The Commission is amending Rule 1000 of Regulation SCI to make competing consolidators exceeding a threshold "SCI entities." See infra Section III.F.

885 See Committee on Capital Markets Regulation Letter at 3; BestEx Research Letter at 1; BlackRock Letter at 5; ACS Execution Services Letter at 5; Clearpool Letter at 7–8; MEMX Letter at 3, 8; ACTIV Financial Letter at 1 (stating that competition could "provide investors with a … higher reliability framework for determining accurate pricing of NMS securities"); NovaSparks Letter at 1 ("The competitive nature of the new model will encourage Competing Consolidators to deliver excellent reliability, functionality and performance.").

886 Committee on Capital Markets Regulation Letter at 3.

887 See BestEx Research Letter at 1. This commenter also said that competing consolidators would reduce reliance on a single SIP vendor. Id. at 5.

888 See BlackRock Letter at 5.

889 ACS Execution Services Letter at 5; Clearpool Letter at 7–8. However, with respect to business continuity planning and disaster recovery, one commenter said that having a single competitor to the SIP would be sufficient to address concerns about a single point of failure. See Data Boiler Letter I at 48.

890 See Cboe Letter at 25; TechNet Letter II at 2. One commenter noted an inverse relationship between data connection speed and reliability. The commenter stated that exchanges and competing consolidators would need to create a protocol for conditional